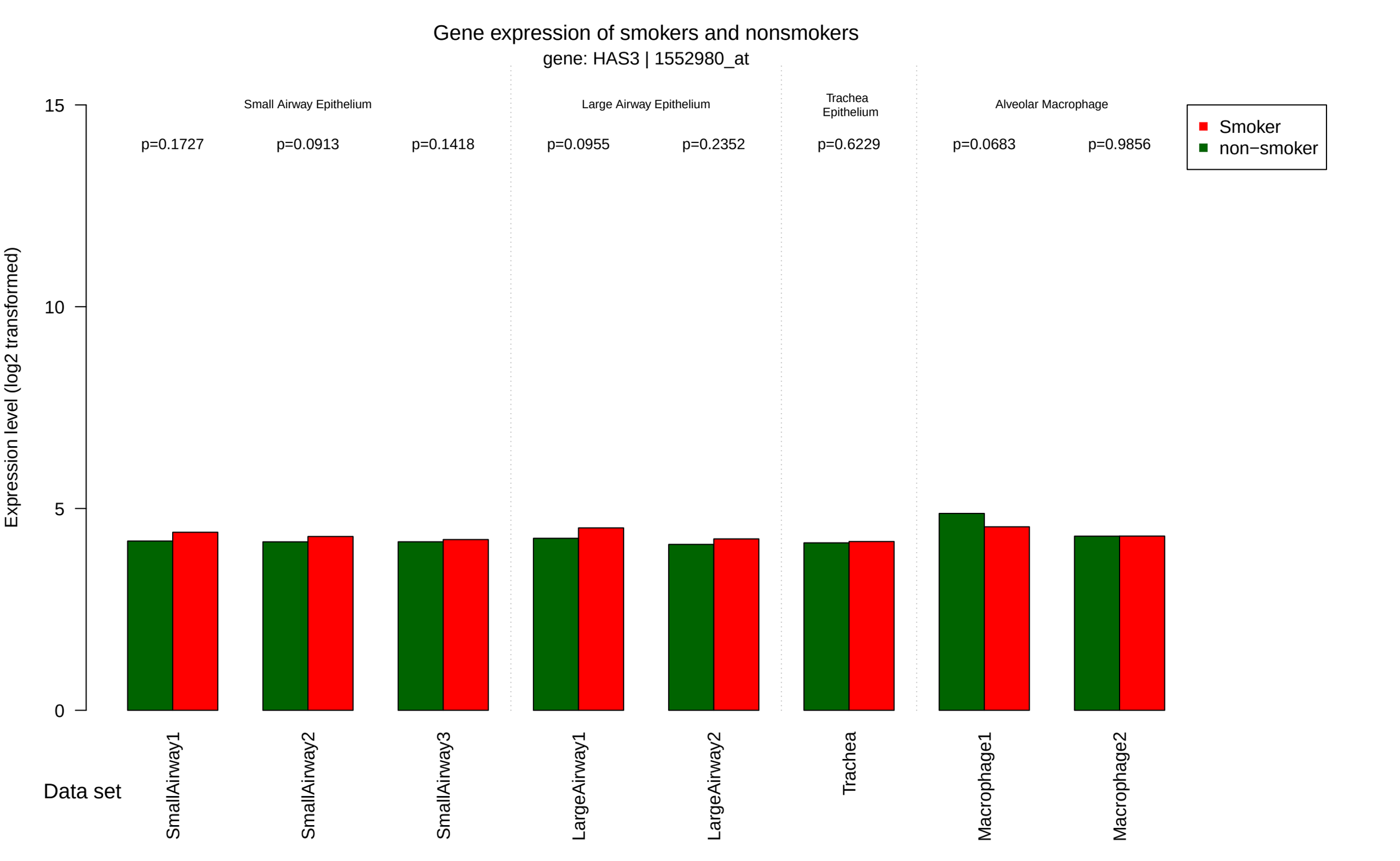
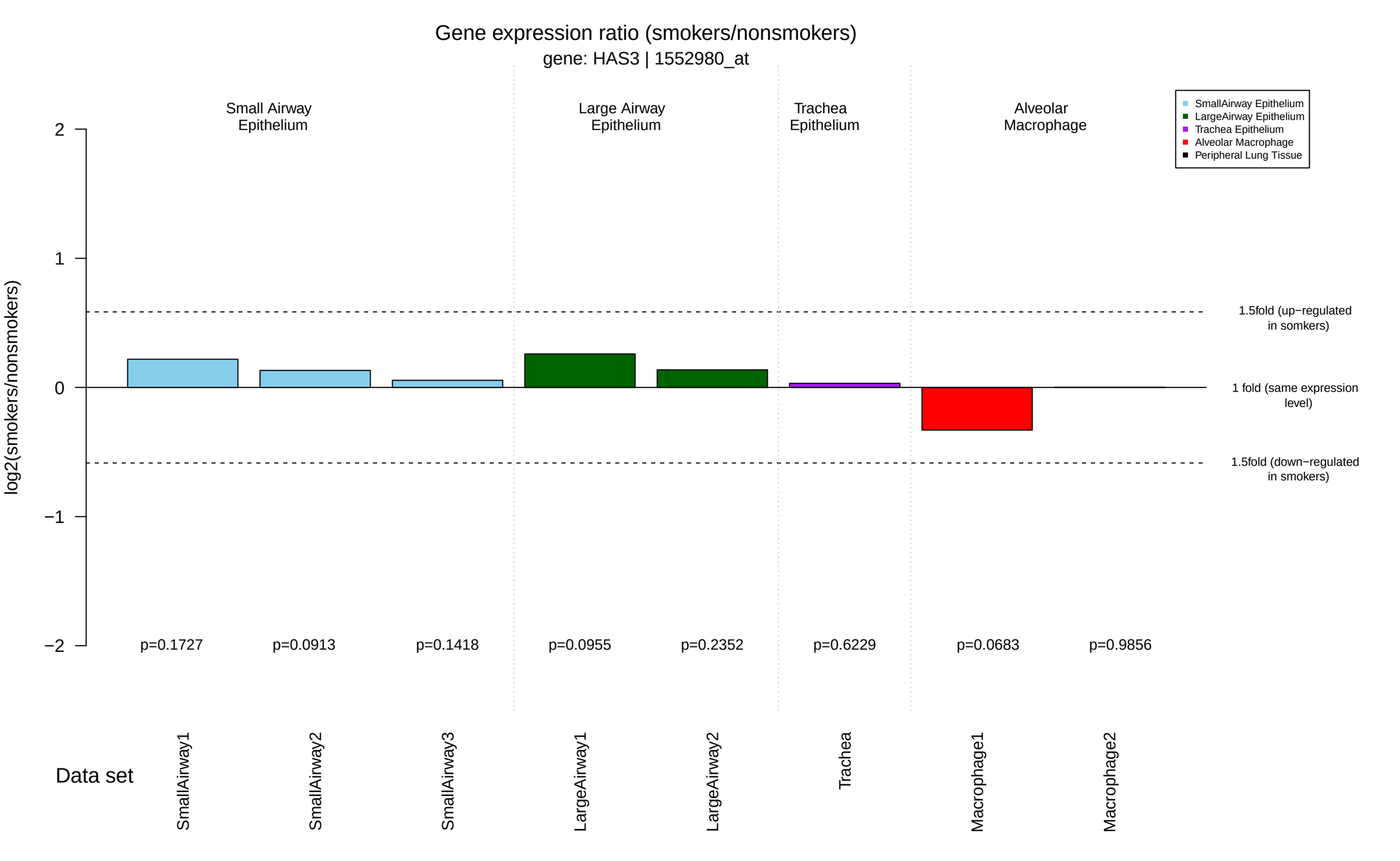
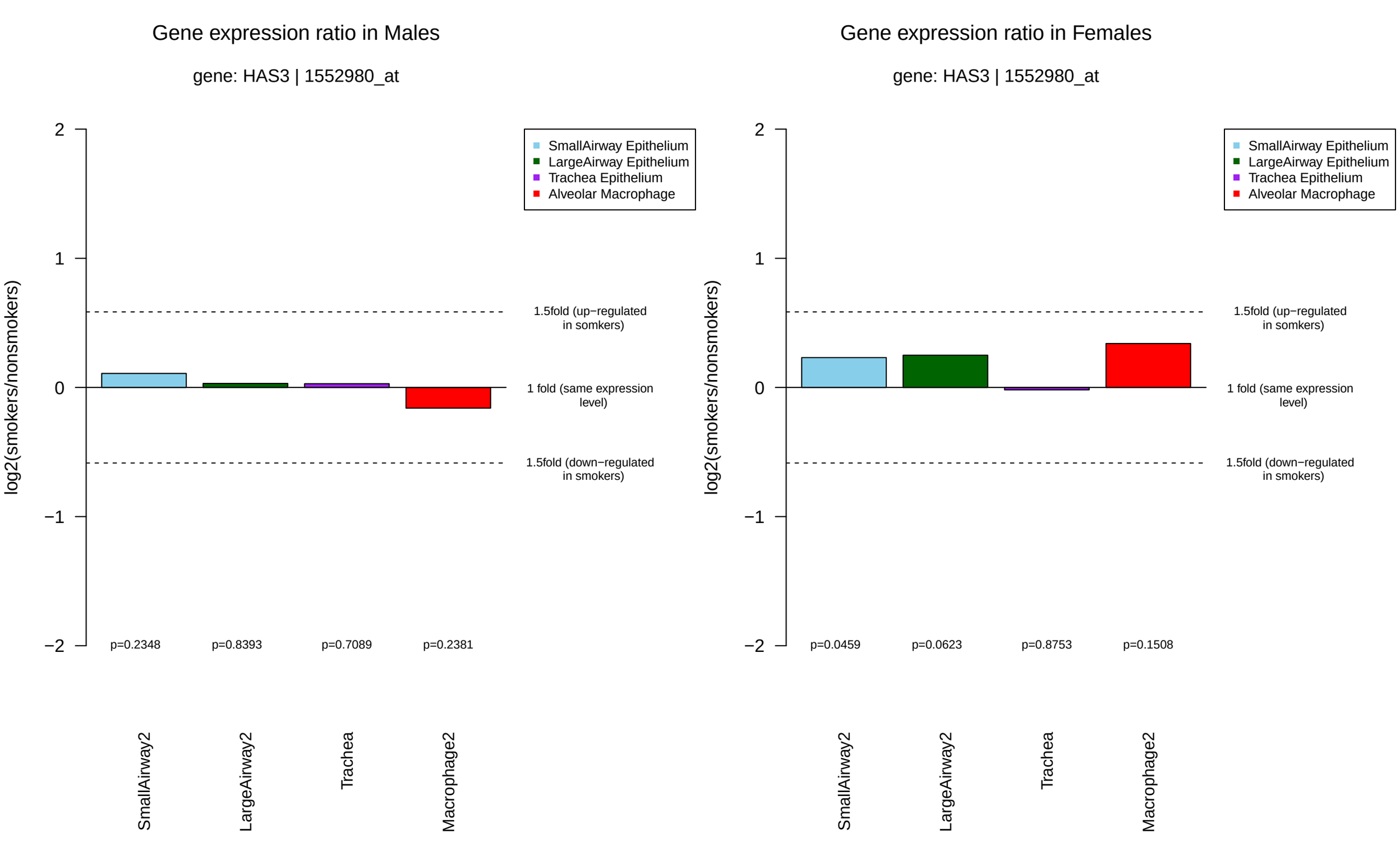
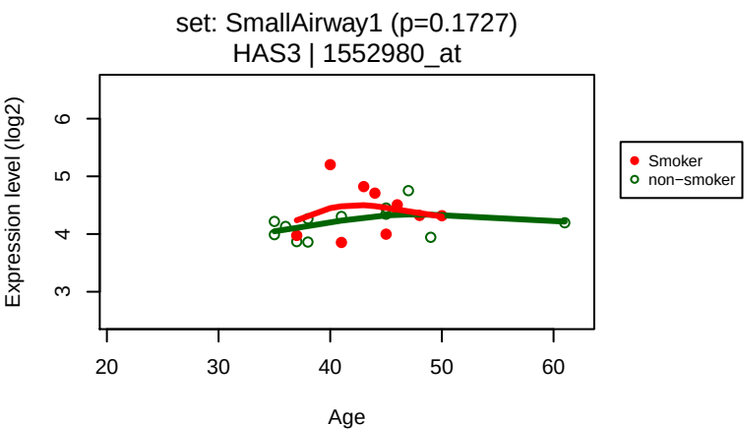
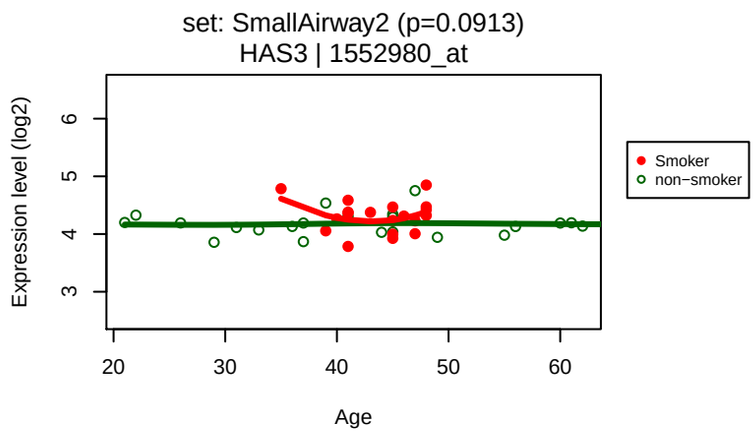
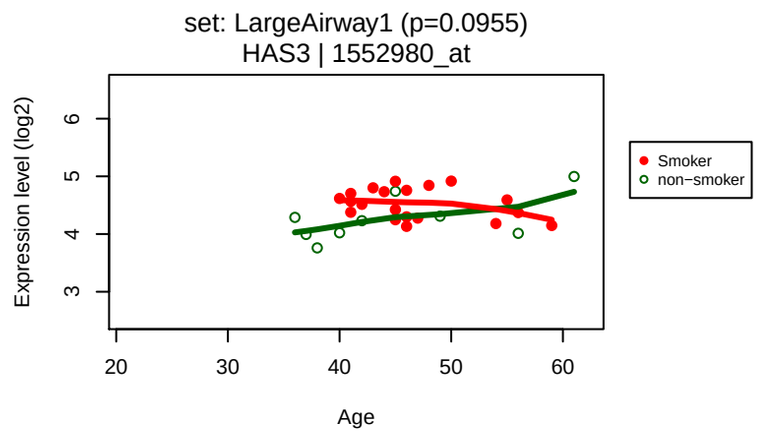

Gene expression of smokers and nonsmokers
gene: HAS3 | 1552980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1727
p=0.0913
p=0.1418
p=0.0955
p=0.2352
p=0.6229
p=0.0683
p=0.9856
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

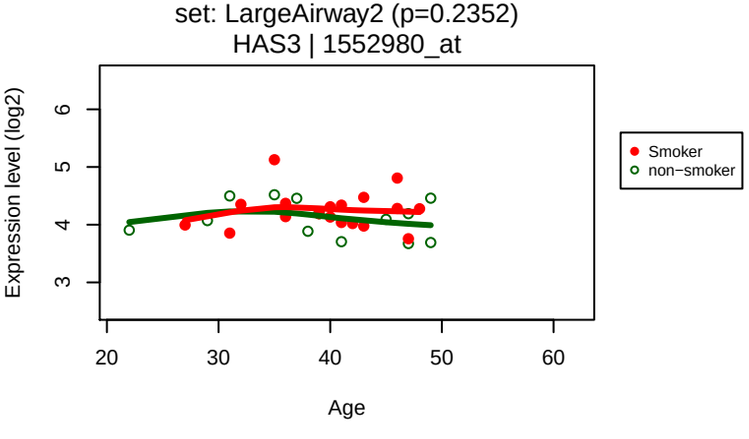

Gene expression ratio (smokers/nonsmokers)
gene: HAS3 | 1552980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1727
p=0.0913
p=0.1418
p=0.0955
p=0.2352
p=0.6229
p=0.0683
p=0.9856
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

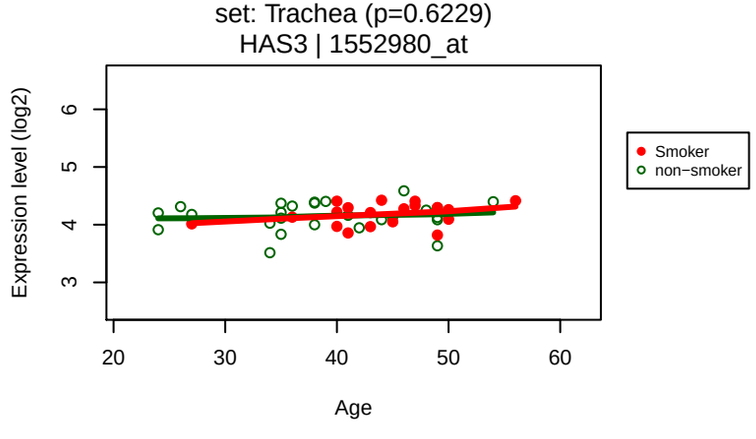

Gene expression ratio in Males
gene: HAS3 | 1552980_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2348
p=0.8393
p=0.7089
p=0.2381
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HAS3 | 1552980_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0459
p=0.0623
p=0.8753
p=0.1508
SmallAirway2
LargeAirway2
Trachea
Macrophage2

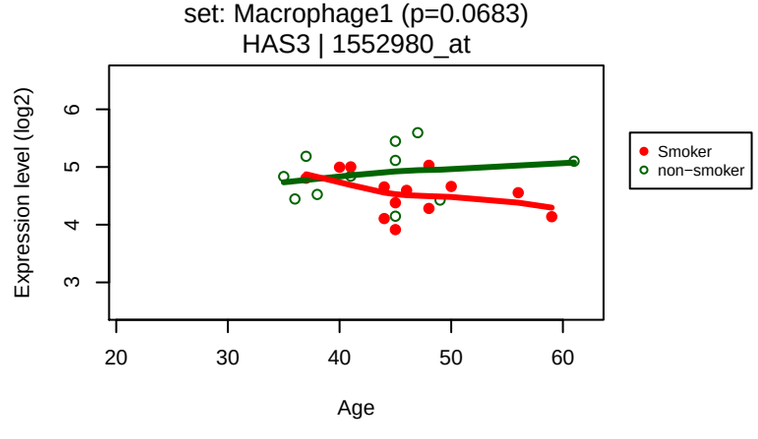

set: SmallAirway1 (p=0.1727)
HAS3 | 1552980_at
Expression level (log2)
Age
Smoker
non−smoker

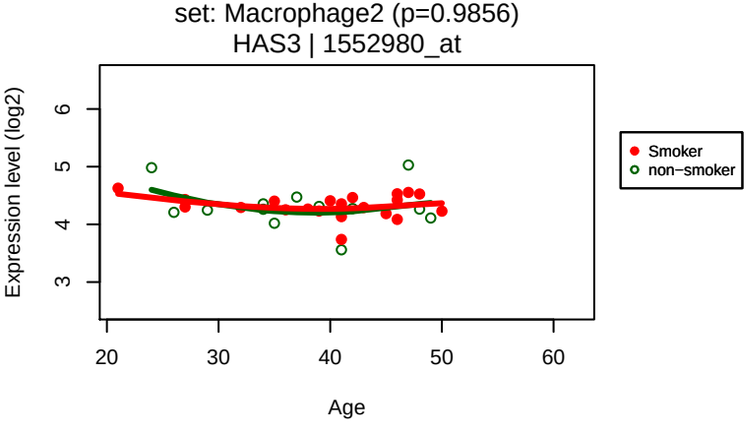

set: SmallAirway2 (p=0.0913)
HAS3 | 1552980_at
Expression level (log2)
Age
Smoker
non−smoker

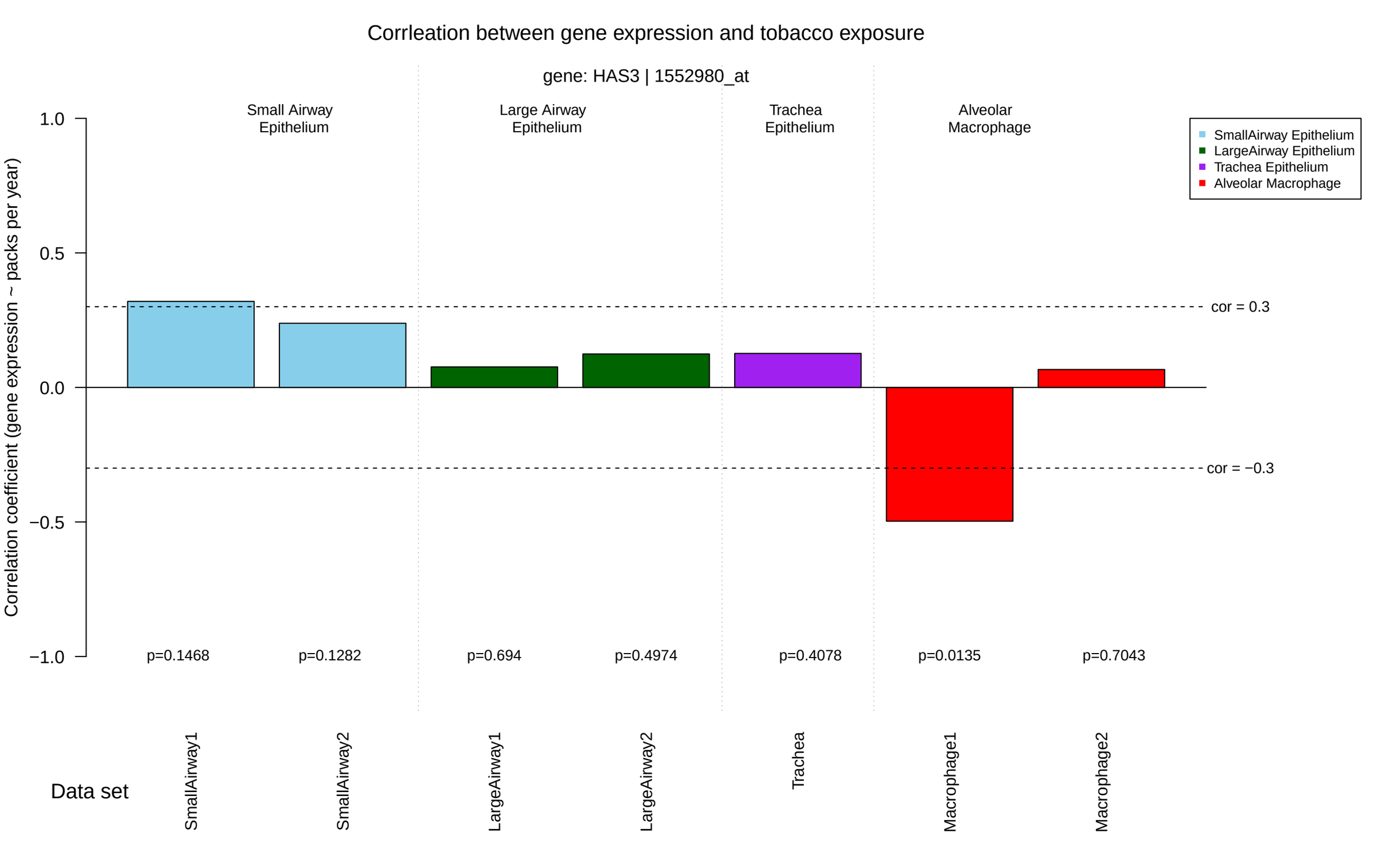

set: LargeAirway1 (p=0.0955)
HAS3 | 1552980_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2352)
HAS3 | 1552980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6229)
HAS3 | 1552980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0683)
HAS3 | 1552980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9856)
HAS3 | 1552980_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HAS3 | 1552980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1468
p=0.1282
p=0.694
p=0.4974
p=0.4078
p=0.0135
p=0.7043
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2